

## Seminar 11

1. Calculati derivatele partiale de ordinul 1, gradientul  $\nabla f$  si diferentia  $df$  pentru functiile

a)  $f : \mathbb{R}^3 \rightarrow \mathbb{R}, \quad f(x, y, z) = x^2 y^3 + y \sin x - 2z$

b)  $f : (0, \infty)^2 \rightarrow \mathbb{R}, \quad f(x, y) = \arctg \frac{x-y}{x+y}$

c)  $f : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad f(x, y) = x\sqrt{x^2 + y^2}$

2. Aratati ca functia  $f(x, y) = y \ln(x^2 - y^2)$  verifica relatia

$$\frac{1}{x} \frac{\partial f}{\partial x} + \frac{1}{y} \frac{\partial f}{\partial y} = \frac{f}{y^2}, \quad \forall x > y > 0$$

3. Studiati existenta derivatelor partiale in origine si a derivatelor dupa directie in origine pentru

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad f(x, y) = \begin{cases} \frac{x^2 y}{x^4 + y^2} & , (x, y) \neq (0, 0) \\ 0 & , (x, y) = (0, 0) \end{cases}$$

4. Calculati derivatele partiale ale functiei compuse  $g \circ f$ , unde  $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2, f(x, y) = (xe^y + xe^{-y}, xe^y - xe^{-y})$  si  $g = g(u, v) : \mathbb{R}^2 \rightarrow \mathbb{R}$  este o functie oarecare de clasa  $C^1$  pe  $\mathbb{R}^2$ .

5. Stiind ca  $g = g(u, v) : \mathbb{R}^2 \rightarrow \mathbb{R}$  este o functie de clasa  $C^1$  cu proprietatea

$$u \frac{\partial g}{\partial v}(u, v) = v \frac{\partial g}{\partial u}(u, v), \quad \forall (u, v) \in \mathbb{R}^2,$$

aratati ca functia  $G(x, y) = g(x, \sqrt{y - x^2})$  satisface relatia

$$\frac{\partial G}{\partial x}(x, y) = 0, \quad \forall (x, y) \in \mathbb{R}^2, y > x^2.$$

6. Exprimati ecuatia

$$u \frac{\partial g}{\partial u}(u, v) + v \frac{\partial g}{\partial v}(u, v) = \sqrt{u^2 + v^2}, \quad \forall (u, v) \in (0, \infty)^2$$

in variabilele  $(x, y) \in (0, \infty) \times (0, \pi/2)$ , efectuand transformarea  $u = x \cos y, v = x \sin y$ . Determinati apoi o functie  $g$  de clasa  $C^1$  ce verifica relatia respectiva.

7. Calculati derivatele partiale de ordinul 2 ale functiilor

a)  $f : (1, \infty) \times \mathbb{R} \rightarrow \mathbb{R}, \quad f(x, y) = \ln(x + y^2 - 1)$

b)  $f : \mathbb{R} \times (0, \infty) \rightarrow \mathbb{R}, \quad f(x, y) = xy e^{\frac{x}{y}}$

## Exercitii suplimentare

1. Calculati derivatele partiale de ordinul 1, gradientul  $\nabla f$  si diferenciala  $df$  pentru functiile

a)  $f : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad f(x, y) = \sin^2(x^3 + y)$

b)  $f : \mathbb{R}^3 \rightarrow \mathbb{R}, \quad f(x, y, z) = (x + y + z)e^{x^2+y^2+z^2}$

2. Calculati matricea Jacobi  $J(f)$  in punctul  $(1, 1)$  pentru urmatoarele functii vectoriale

a)  $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3, \quad f(x, y) = (x^2 - y, 3x - 2y, 2xy + y^2)$

b)  $f : (0, \infty)^2 \rightarrow \mathbb{R}^2, \quad f(x, y) = (\frac{1}{xy}, \arctg \frac{y}{x})$

**Functii omogene.** Fie  $p \in \mathbb{R}$ . O functie  $f : (0, \infty)^m \rightarrow \mathbb{R}$  se numeste omogena (de grad  $p$ ) daca  $f(tx) = t^p f(x), \forall x \in (0, \infty)^m$  si  $\forall t > 0$ .

3. Aratati ca functia  $f : (0, \infty)^3 \rightarrow \mathbb{R}, f(x, y, z) = \frac{1}{x+y+z} \left( \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right)$  este omogena (de un anumit grad) si justificati egalitatea

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + z \frac{\partial f}{\partial z} = -2f, \quad \forall (x, y, z) \in (0, \infty)^3$$

4. Aratati ca functia de mai jos nu este continua in  $(0, 0)$ , dar admite derivate dupa orice directie in acest punct

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad f(x, y) = \begin{cases} \frac{xy}{x+y}, & x+y \neq 0 \\ 0, & x+y = 0 \end{cases}$$

5. Calculati derivatele partiale ale functiei compuse  $g \circ f$ , unde  $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3, f(x, y) = (x^2 - y, 3x - 2y, 2xy + y^2)$  si  $g = g(u, v, w) : \mathbb{R}^3 \rightarrow \mathbb{R}$  este o functie oarecare de clasa  $C^1$  pe  $\mathbb{R}^3$ .

6. Exprimati ecuatia

$$v \frac{\partial g}{\partial u}(u, v) - u \frac{\partial g}{\partial v}(u, v) = 1, \quad \forall (u, v) \in (0, \infty)^2$$

in variabilele  $(x, y) \in (0, \infty) \times (0, \pi/2)$ , efectuand transformarea  $u = x \cos y, v = x \sin y$ . Determinati apoi o functie  $g$  de clasa  $C^1$  ce verifica relatia respectiva.

7. Aratati ca functia  $f(x, y) = (x^2 + y^2) \arctg \frac{y}{x}$  verifica relatia

$$x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = 2f, \quad \forall (x, y) \in (0, \infty)^2$$